

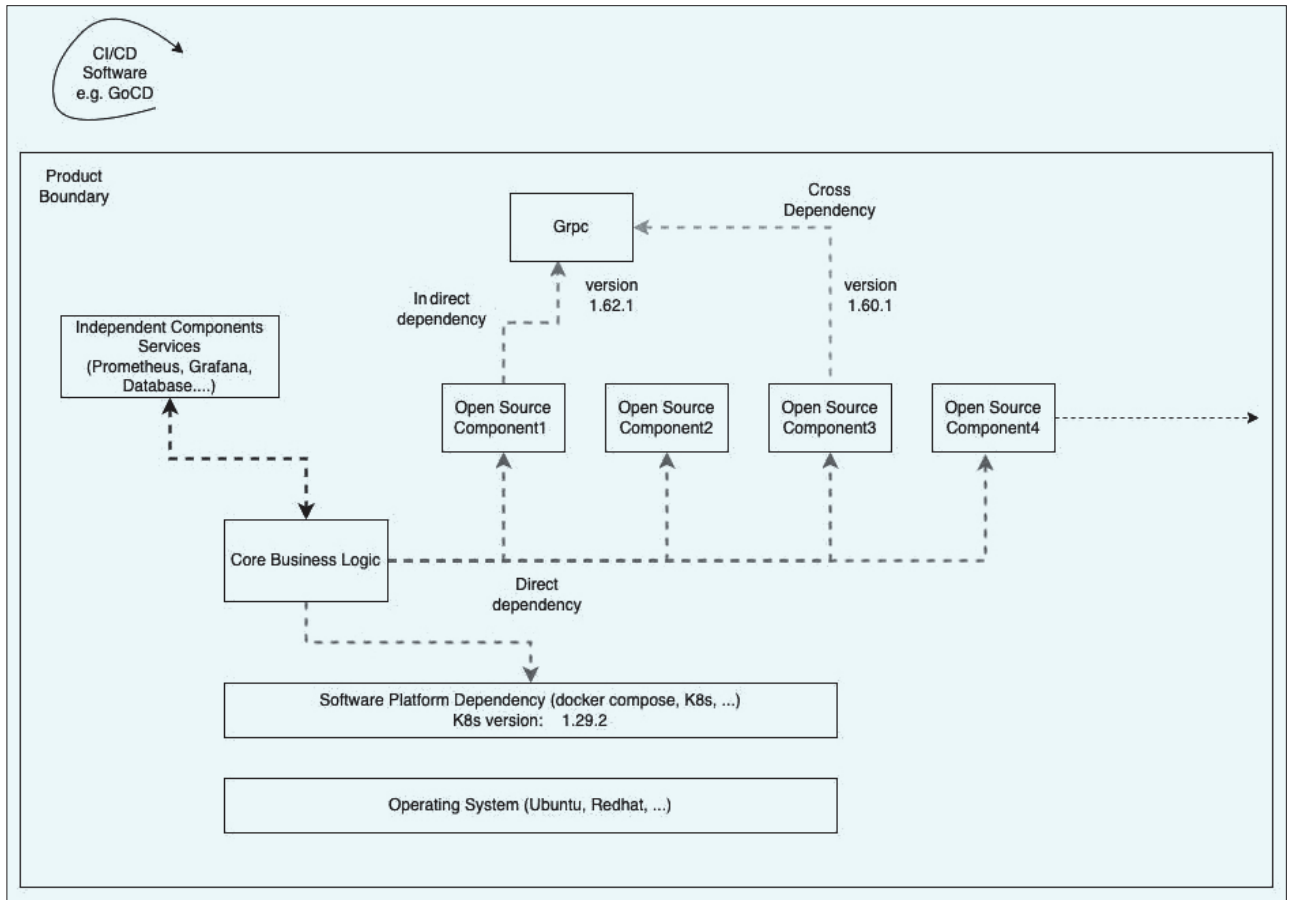


Figure 1: Dependency diagram

- **Deployment subsystem:** Many times, complex projects are deployed using the CI/CD solution. Though CI/CD does not use the code directly, it controls the way the project is deployed. Project Flux is a simple example of how a project could be deployed.

Except for the core business logic, all the blocks in Figure 1 are open source. Imagine that each block has its own release cycle and life span. Now we need to ensure that our product can be reliably serviced and upgraded while not failing to meet any of our client SLAs. Remember, this is not a one-time operation.

Reasons for open source component upgrades

After completing a project, an organisation should keep it in good

working order. As time passes, the project must accept upgrades from the open source components. This is always necessary for a variety of reasons, and if not completed, the project becomes insecure from a security standpoint and unmaintainable from a functional standpoint. It is vital to understand that updates are ongoing activities. This is a significant task for a company wishing to employ open source projects. The following are some reasons why open source components should be upgraded.

Upgrades required due to end of life (EoL)

Sometimes project module version changes are necessary simply because some components have reached the end of their support life cycle. This must be

arranged correctly and depends on the sort of component that must be updated. The work necessary to update one library may be far less than that required to change a whole software platform. It is crucial to highlight that most well-governed projects have a clear release schedule and support requirements.

The website <https://endoflife.date/> offers timetables for numerous software projects' end of life dates. This is a useful resource for many product managers. Some projects have a well-defined release cycle and end of support. For example, Kubernetes, a leading container orchestration technology, has released information on their release cycles and how long software is maintained at <https://kubernetes.io/releases/release/>.

It can be seen in Figure 1 that a Kubernetes update is critical and